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The Laws That Control the Universe

Mapping the World and Deciphering the Laws of Physics

Physicists have struggled from the seventeenth century onward to discover the physical laws that shape and control our universe. This has been like European explorers struggling to discover the Earth's geography (Figure 3.1).

By 1506 Eurasia was coming into focus and there were glimmers of South America. By 1570 the Americas were coming into focus, but there was no sign of Australia. By 1744 Australia was coming into focus, but Antarctica was terra incognita.

Similarly (Figure 3.2), by 1690 the *Newtonian laws of physics* had come into focus. With concepts such as force, mass, and acceleration and equations that link them, such as $F = ma$, the Newtonian laws accurately describe the motion of the Moon around the Earth and the Earth around the Sun, the flight of an airplane, the construction of a